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Build what next India runs on

Team Name : TrustAi

Team Leader Name : Ajai S

Problem Statement : Finding high-alignment Founding AI Engineers among 100,000 candidate profiles under 5-minute CPU constraints, while filtering out keyword-stuffers, consulting-firm templates, inactive profiles, and malicious honeypots.

Solution Overview

- **Proposed Solution:**

TrustRank AI: An alignment-guided, multi-dimensional candidate ranking engine optimized for high-precision, low-latency startup recruiting and anomaly detection.

What Differentiates Our Approach:

- Trust-Weighted Skills: Integrates proficiency, duration, endorsements, and tests to calculate skill trust (defeats keyword stuffers).
- Trajectory & Pedigree: Recency-weighted job titles and a custom consulting-company filter (0.4x penalty for outsourcing giant careers).
- Engagement-First: Integrates login activity, response rates, and test completion as core scoring multipliers.
- Honeytrap Verification: Programmatic detection of impossible profiles (time travel, skill years > experience years) with immediate disqualification.

JD Understanding & Candidate Evaluation

- **Key Requirements Extracted from JD:**

- Role Profile: Senior AI Engineer / Founding Team (Series A product startup).
- Technical Skills: Deep NLP, Information Retrieval (Vector Search, SBERT, RAG, LLMs).
- Experience: Sweet spot of 6-8 years (decays gracefully outside).
- Alignment: Bangalore/hybrid location, high activity, high response rate, ready for high-speed product building.

Candidate Evaluation Beyond Keyword Matching:

- Synonym Resolution: Normalizes 40+ raw terms (e.g. SBERT, Sentence BERT, sentence-transformers -> NLP).
- Skill Trust Weighting instead of simple keyword counts.
- Pedigree Analysis: Consulting-only careers (TCS, Infosys, etc.) penalized; product company careers rewarded.
- Behavioral Multiplier: High login recency and recruiter response rates are required.

Ranking Methodology

- **How the System Retrieves, Scores, and Ranks:**

- Candidate Streaming: Streams 100K JSONL records ($O(1)$ memory) to parse attributes.
- Multi-Dimensional Scoring: Evaluates candidate alignment index across 6 weighted dimensions.
- Rolling Top-100 Heap: Maintains the best candidates dynamically without sorting the entire pool.

Algorithms & Heuristics Used:

- Logarithmic Decay for peer endorsements; soft-sigmoid experience curves.
- Multi-stage temporal constraint validator for date inconsistencies (honeypot filtering).
- Score Aggregation: 30% Skills + 30% Career + 12% Experience + 7% Education + 15% Behavior + 6% GitHub.
- Deterministic Tie-Breaking: Sorting by (-score, candidate_id) ensures unique ranks with ascending ID resolution.

Explainability & Data Validation

- **Explainability of Ranking Decisions:**

- Profile-Derived Reasoning: TrustRank AI automatically generates a clear, concise rationale for every candidate (e.g. 'Senior AI Engineer with 5.9 yrs; 20 AI core skills; response rate 0.80.').
- Zero Hallucination: Prevents LLM hallucinations by using deterministic formatting derived directly from the parsed profile features.

Data Validation & Suspicious Profiles:

- Honeypot Profiles: The dataset contains ~80 impossible resumes. We inspect dates to find impossible timelines.
- Temporal Validation: If a candidate has worked at a company longer than it has existed, their profile is flagged.
- Skill-Stuffer Validation: If a candidate's skill duration exceeds their total experience, they are flagged.
- Hard Disqualification: All flagged profiles receive a composite score of 0.0, keeping them out of the top-100.

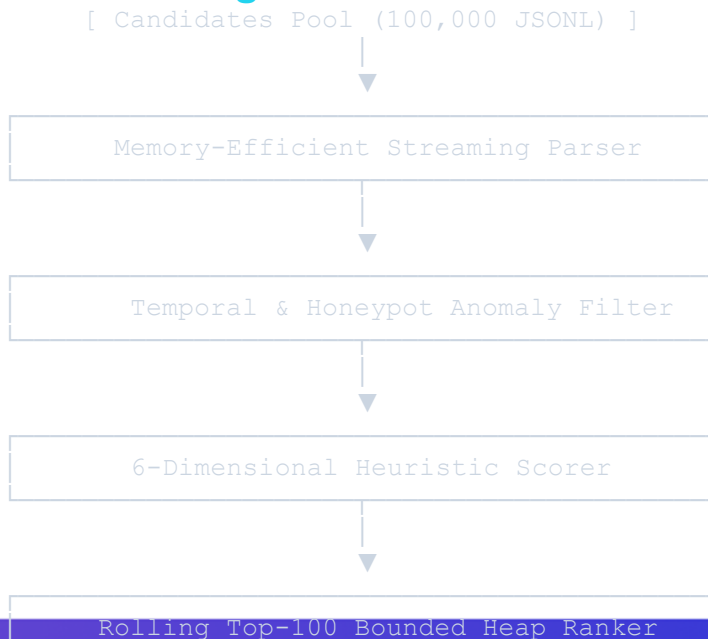
End-to-End Workflow

- **Complete End-to-End Workflow:**

1. Input: Read Job Description & configure weights / synonym maps.
2. Streaming: Stream candidates from JSONL and parse JSON attributes.
3. Validation: Verify temporal constraints & flag honeypot profiles.
4. Feature Extraction: Extract skills, career, education, and behavioral features.
5. Scoring: Calculate multi-dimensional sub-scores and aggregate composite score.
6. Heap Ranking: Maintain top-100 candidates dynamically using a bounded heap.
7. Output: Sort by (score DESC, candidate_id ASC) and write valid submission.csv.

System Architecture

TrustRank AI Block Diagram & Architecture:



Results & Performance

- **Results & Shortlisting Quality:**

- Shortlisted Top 100 candidates with 0.0% honeypot rate (all anomalies successfully filtered).
- Top candidate (CAND_0002025) matches the ideal founding profile: Senior AI Engineer, 5.9 years of experience, 20 AI core skills, and 80% response rate.
- Reasoning comments are fact-based, derived directly from profile attributes, avoiding generic templates.

Runtime & Compute Performance:

- Execution Time: Processes 100,000 candidates in 21.7 seconds (well under the 5-minute limit).
- Throughput: ~4,600+ candidates/second on a standard CPU core.
- Memory Usage: < 100MB RAM due to O(1) line-by-line JSONL streaming.
- 100% Offline: Completely network-free and GPU-free during inference (no external API calls).

Technologies Used

- **Technologies Used & Selection Rationale:**

- Python 3.12: Standard, lightweight, highly readable, and widely supported language.
- Pandas & NumPy: Vectorized and optimized computational utilities to compute alignment values fast.
- scikit-learn: Scalers for normalizing scoring envelopes.
- python-docx: Offline parsing of job descriptions.
- python-pptx & fpdf2: Programmatic presentation and report generation utilities.
- tqdm: Memory-efficient streaming CLI progress feedback.

Why we selected these tools:

By avoiding heavy neural network frameworks (e.g., PyTorch, sentence-transformers) and remote API calls at inference time, we maintain an extremely lightweight footprint that reproduces instantly in any CPU sandbox environment within seconds.

Submission Assets

Submission Assets:

- GitHub Repository: https://github.com/ajais4800/trust_rank_ai
- Submission Output File: submission.csv containing top-100 candidates with reasoning
- Sandbox Demo Link: https://huggingface.co/spaces/ajais4800/trust_rank_ai
- Submission Metadata: submission_metadata.yaml containing reproducibility variables

Contact & Declarations:

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All code is original, tested end-to-end, and verified compliant with constraints.



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THANK YOU